UPLIFT MODELING FOR PERSONALIZED MARKETING IN TELECOMMUNICATIONS

Applied Machine Learning, final project

Abilkhay Askar, Kolzhabay Olzhas

Group Big Data Analysis 2406, Astana IT University

CONTENTS

01	**Introduction**	Problem, why response models are not enough, research question	slides 3–4
02	**Data**	MegaFon dataset, quality checks, treatment and control balance	slides 5–6
03	**Methodology**	Our workflow, data pipeline, model architecture, how a customer is scored	slides 7–12
04	**Experiments / Results**	Metrics table, Qini curves, five splits, deciles, campaign effect	slides 13–18
05	**Discussion**	Why the T-learner won, overfitting, limitations	slides 19
06	**Conclusion**	Main findings and next steps	slides 20
07	**References**	Literature and tools	slides 21

Structure follows the final project requirements: Introduction, Data, Methodology, Experiments/Results, Discussion, Conclusion, References

WHY RESPONSE MODELS ARE NOT ENOUGH

Usual practice

A campaign is sent to customers with the highest predicted probability to buy. Many of them would buy without the offer, so the money is spent but sales do not grow.

Uplift modeling

We estimate how much the offer changes the probability to buy for each customer:

$$\text{uplift}(x) = P(Y=1 \mid x, T=1) - P(Y=1 \mid x, T=0)$$

This needs a randomized experiment: one group got the offer (T = 1), the other did not (T = 0).

	WITH THE OFFER: BUYS	WITH THE OFFER: DOES NOT BUY
WITHOUT THE OFFER: BUYS	**Sure things** Buy anyway. The offer is wasted.	**Sleeping dogs** The offer pushes them away. Do not contact.
WITHOUT THE OFFER: DOES NOT BUY	**Persuadables** Buy only because of the offer. Target them.	**Lost causes** Never buy. The offer is wasted.

Customer types: Radcliffe (2007); uplift definition: Gutierrez & Gérardy (2017)

RESEARCH QUESTION

Which subscribers should receive a marketing offer to get the most additional conversions, and does an uplift model rank them better than random targeting and a normal conversion model?

Objectives

1. Check that the experiment is randomized, so uplift can be estimated
2. Build at least two uplift models and compare them with two baselines
3. Evaluate with uplift metrics (Qini, AUUC, uplift@30%) on unseen data
4. Turn the best model into a list of customers to contact

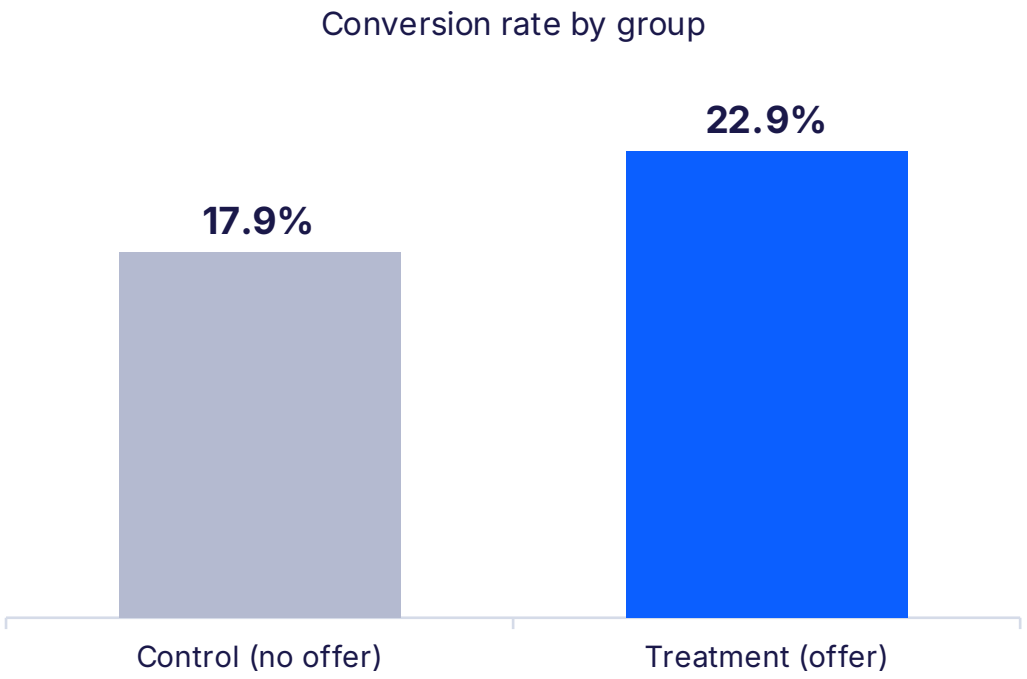
Why it matters for a telecom operator

- Every contact (SMS, call, bonus) costs money
- Operators already keep random control groups
- Millions of subscribers: a small gain in targeting gives thousands of extra conversions
- Some customers react badly to offers

THE MEGAFON DATASET

Data

PROPERTY	VALUE
Source	MegaFon Uplift Competition, ODS.ai (2021)
Type	Synthetic data that imitates a telecom campaign
Rows	600,000 labelled customers
Features	50 numeric, anonymized (X_1 to X_50)
Treatment	treatment_group: offer sent (300,368) or control (299,632)
Target	conversion, 1 if the customer converted
Missing / duplicates	0 / 0



+4.95 pp

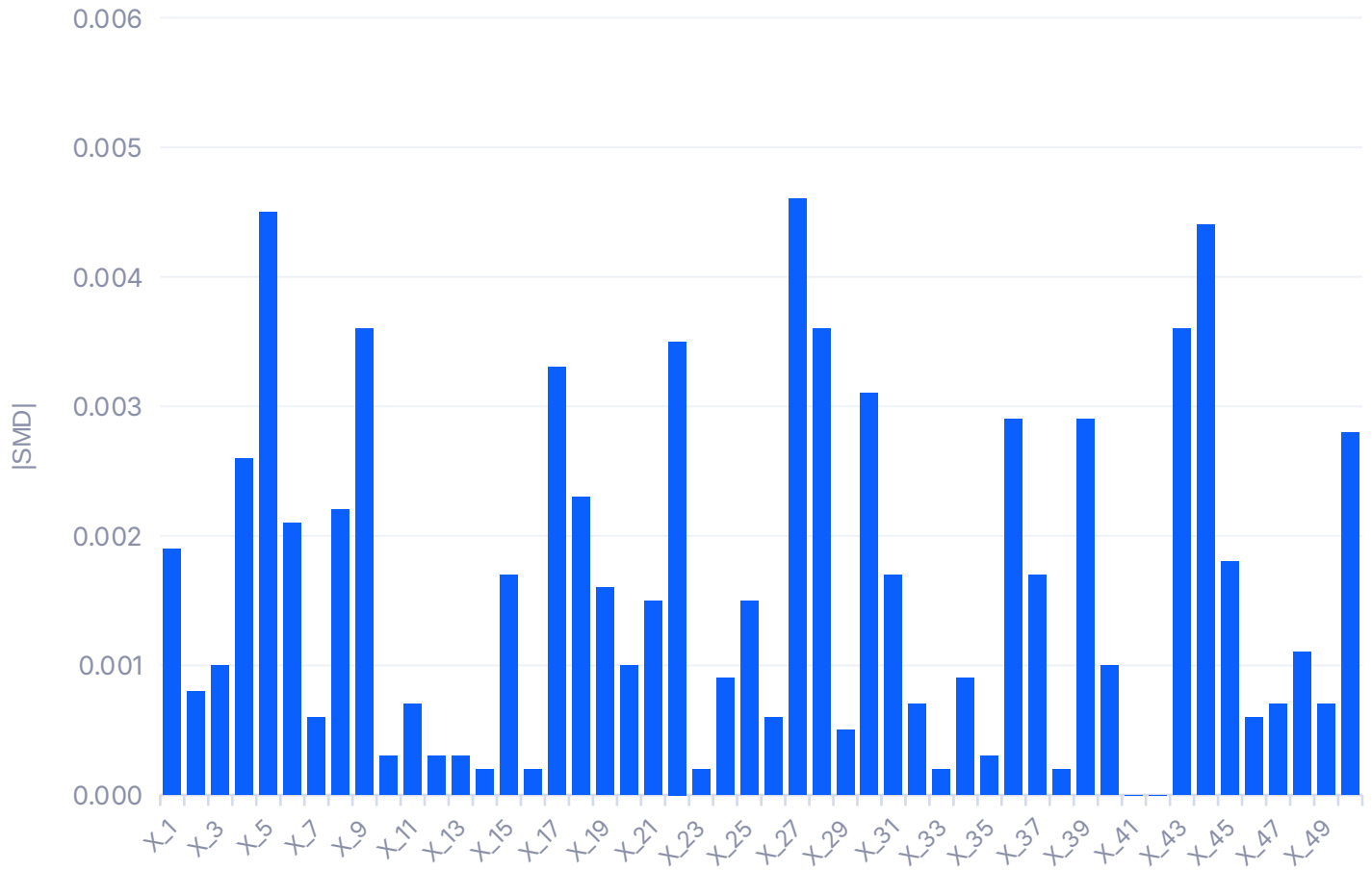
average effect of the offer (95% CI 4.75 to 5.16 pp), a 27.6% relative lift

MegaFon Uplift Competition (ODS.ai, 2021), distributed with scikit-uplift; 95% CI for the effect from a two-proportion normal approximation

DATA QUALITY AND GROUP BALANCE

Data

Standardized mean difference (SMD) between treatment and control for every feature



CHECK	RESULT
Missing values	0
Duplicate rows	0
Max |SMD|	0.0046 (X_27)
Features with |SMD| ≥ 0.1	0 of 50
Train / test rows	420,000 / 180,000
Treatment share, train / test	50.06% / 50.06%
Conversion, train / test	20.42% / 20.42%

All values are far below 0.1, so the two groups are comparable and differences in conversion come from the offer.

SMD = (mean_T − mean_C) / sqrt((var_T + var_C) / 2); usual imbalance threshold 0.1; split stratified on treatment × conversion, seed 42

OUR WORKFLOW

Methodology

How we did the project, from the proposal to the final targeting list

1 **Proposal**
Chose the dataset, metrics and 9 success criteria

2 **Data checks**
Missing values, groups, balance of 50 features

3 **Split**
70 / 30, stratified by treatment and conversion

4 **Baselines**
Random scores and a conversion classifier

5 **Uplift models**
S-learner, T-learner, class transformation

6 **Tuning**
4 settings each, 3-fold CV Qini on train

7 **Evaluation**
Test set metrics and 5 repeated splits

8 **Targeting**
Ranked list, top 30% get the offer

Tools

Google Colab, Python 3.13, pandas, scikit-learn 1.9, scikit-uplift 0.5.1, matplotlib

Rules we kept

The test set is never used for tuning. Models are chosen by Qini, not by accuracy. Every result is repeated on 5 splits.

Outputs

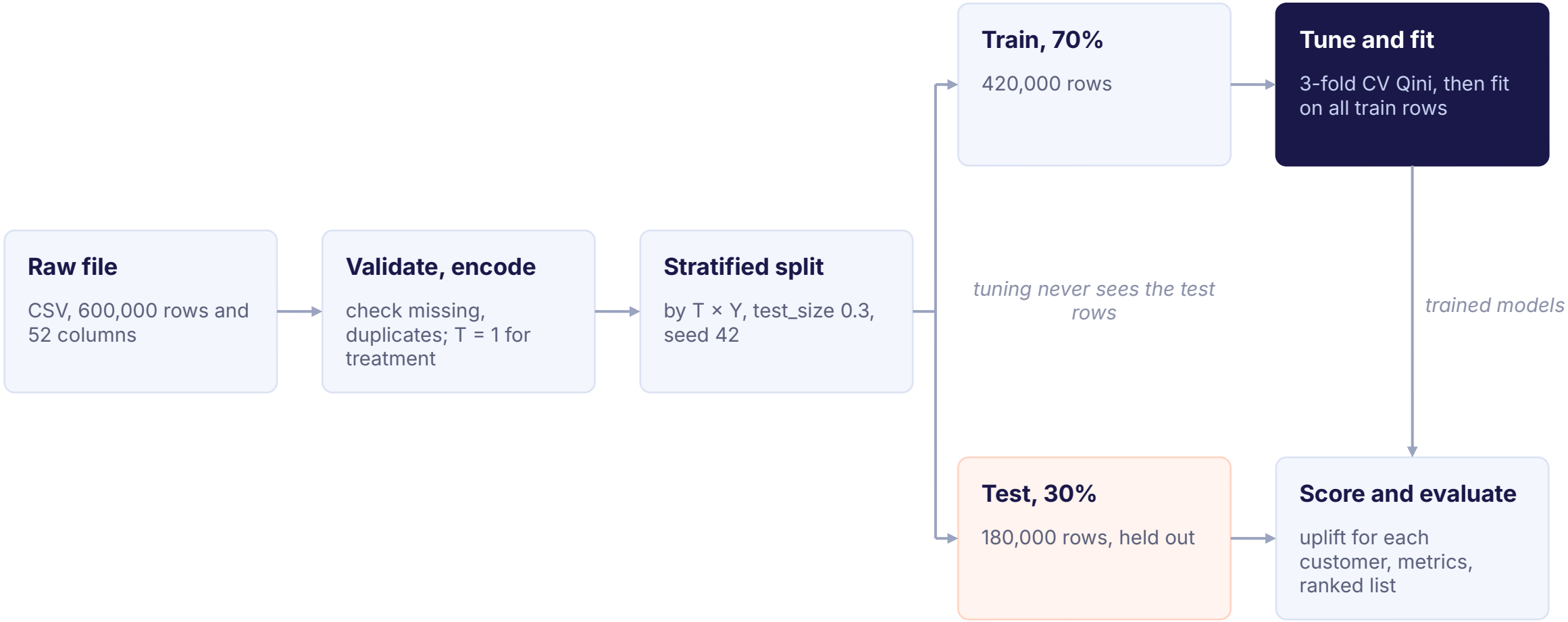
Executed notebook, figures and tables, targeting_list_top30.csv, research paper, these slides

Everything runs in one Google Colab notebook; all random seeds fixed; full run took 14 minutes on 2 CPU cores

DATA PIPELINE

Methodology

The test set is split off once and only used at the end



Notebook sections 2 to 11; stratification keeps the treatment share and conversion rate identical in both parts

MODEL ARCHITECTURE: T-LEARNER

Methodology

1. TRAINING (TRAIN SET ONLY)

Training set

420,000 customers with X, T and Y

Treated, T = 1

about 210,000 rows

Control, T = 0

about 210,000 rows

Model f_1

gradient boosting, learns P(buy | offer)

Model f_o

same settings, learns P(buy | no offer)

Each model sees only its own group, so the two can learn different patterns.

2. PREDICTION (EVERY CUSTOMER)

Customer x

50 features, treatment flag not needed

$f_1(x) = 0.90$

chance to buy with the offer

$f_o(x) = 0.07$

chance to buy without it

uplift(x) = $f_1(x) - f_o(x) = 0.83$

extra probability created by the offer

Rank all customers by uplift

top 30% get the offer, negative uplift are left out

TwoModels(method='vanilla') from scikit-uplift; base learner HistGradientBoostingClassifier: 150 trees, 63 leaves, learning rate 0.1, min_samples_leaf 100, L2 1.0

HOW THE MODEL SCORES A CUSTOMER

Methodology

Three real customers from the test set, scored by the final T-learner

CUSTOMER	P(BUY | OFFER)	P(BUY | NO OFFER)	UPLIFT	RANK	DECISION
435517	**0.90**	**0.07**	**+0.83**	**1**	**Target**
492666	0.21	0.10	+0.11	54,000	Target
519801	0.05	0.96	−0.91	180,000	Do not target

Rank 1: the offer adds 83 points of probability, the best customer to contact.

Rank 54,000: the last customer inside the 30% budget (54,000 of 180,000).

Rank 180,000: would buy without the offer (0.96) but probably not with it. A response model would contact this customer first.

notebook, sections 7 and 11

```
params = dict(lr=0.1, leaves=63,
              iters=150)
model = TwoModels(
    estimator_trmnt=hgb(**params),
    estimator_ctrl=hgb(**params),
    method="vanilla")
model.fit(X_tr, y_tr, t_tr)

uplift = model.predict(X_te)
p1 = model.trmnt_preds_  # offer
p0 = model.ctrl_preds_   # no offer
```

hgb() = HistGradientBoostingClassifier with min_samples_leaf 100, L2 1.0, seed 42

results/targeting_list_top30.csv from the notebook; probabilities rounded to two decimals

MODELS WE COMPARED

Methodology

Three uplift approaches, one linear reference and two baselines

APPROACH	HOW IT IS TRAINED	SCORE USED FOR RANKING	CLASS
S-learner	one model on X plus the flag T	f(x, 1) − f(x, 0)	SoloModel
T-learner	**two models: treated rows and control rows**	**$f_1(x) - f_o(x)$**	**TwoModels**
Class transformation	one model on Z = 1 when T equals Y	$2 \cdot P(Z=1 \mid x) - 1$	ClassTransformation
Linear T-learner	T-learner with logistic regression	$f_1(x) - f_o(x)$	TwoModels
Conversion model	one classifier on X, ignores T (baseline)	$P(Y=1 \mid x)$	scikit-learn
Random	no training (baseline)	random number	numpy

Class transformation works only because the treatment share is close to 50%, which is true here (50.06%).

Künzel et al. (2019); Jaskowski & Jaroszewicz (2012); all tree models use HistGradientBoostingClassifier (Pedregosa et al., 2011)

TRAINING SETUP AND METRICS

Methodology

Settings were chosen by 3-fold cross-validated Qini on the training set

CV QINI BY SETTING

lr / leaves / trees	S-learner	T-learner	Class transf.
0.10 / 31 / 200	0.2313	0.2364	0.2184
0.05 / 31 / 300	0.2310	0.2361	**0.2233**
0.10 / 15 / 200	0.2179	0.2292	0.2177
0.10 / 63 / 150	**0.2365**	**0.2380**	0.2179

Selected settings are highlighted. More leaves helped the S- and T-learner; class transformation preferred a lower learning rate with more trees.

Qini coefficient

Area between the model's Qini curve and the random line, divided by the area of a perfect model. Random = 0, perfect = 1. Main metric.

AUUC

The same normalization on the uplift curve.

Uplift@30%

Treatment conversion minus control conversion among the 30% of customers with the highest scores.

Why not accuracy

A model can predict who buys and still rank people badly by the effect of the offer.

Grid of 4 settings per learner (learning rate / max leaves / trees); fixed: min_samples_leaf 100, L2 1.0, no early stopping, seed 42

RESULTS ON THE TEST SET

180,000 customers that no model saw during training or tuning

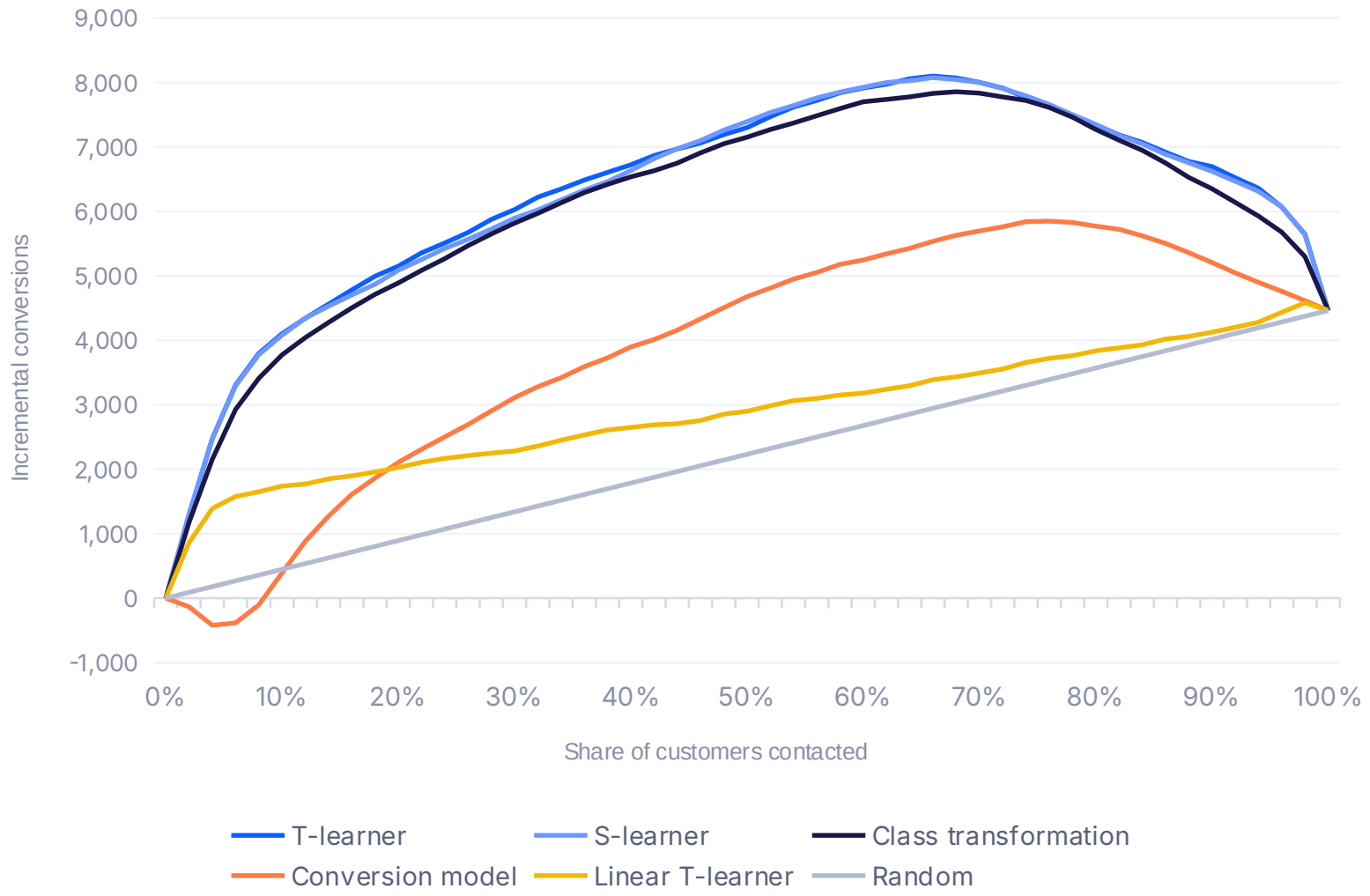
MODEL	QINI	AUUC	UPLIFT@30%	MEAN, 5 SPLITS	SD
T-learner	**0.245**	0.170	22.4%	0.244	0.0026
S-learner	**0.243**	0.168	21.9%	0.241	0.0046
Class transformation	**0.231**	0.160	21.7%	0.230	0.0027
Conversion model	**0.096**	0.067	11.5%	0.096	0.0026
Linear T-learner	**0.041**	0.028	8.5%	0.036	0.0030
Random	**−0.003**	−0.002	4.5%	−0.002	0.0034

The T-learner ranks customers best (Qini 0.245). That is 2.5 times the conversion model (0.096), which targets customers who are simply likely to buy.

Qini and AUUC normalized (random = 0, perfect = 1); overall observed uplift on the test set 0.0495; last two columns: mean and SD over 5 splits

QINI CURVES

Incremental conversions as more customers are contacted, in order of each model's score



What we see

The three boosted uplift models almost overlap and peak at about 8,000 extra conversions near 65% of customers.

After that the curves fall, because the last customers buy less when they get the offer.

The conversion model starts below zero: its top customers are often the ones the offer pushes away.

Held-out test set, 180,000 customers; curves sampled every 2% of customers; grey line = random targeting

STABILITY OVER FIVE SPLITS

Qini coefficient on the test part of five different stratified 70 / 30 splits

MODEL	SEED 42	SEED 7	SEED 123	SEED 2024	SEED 99	MEAN	SD
T-learner	**0.245**	**0.247**	**0.242**	**0.240**	**0.245**	**0.244**	**0.0026**
S-learner	0.243	0.245	0.241	0.233	0.242	0.241	0.0046
Class transformation	0.232	0.234	0.231	0.226	0.229	0.230	0.0027
Conversion model	0.097	0.092	0.095	0.098	0.098	0.096	0.0026
Linear T-learner	0.041	0.033	0.036	0.035	0.034	0.036	0.0030
Random	−0.003	−0.007	−0.000	−0.002	0.002	−0.002	0.0034

5 of 5 splits where the T-learner beats random

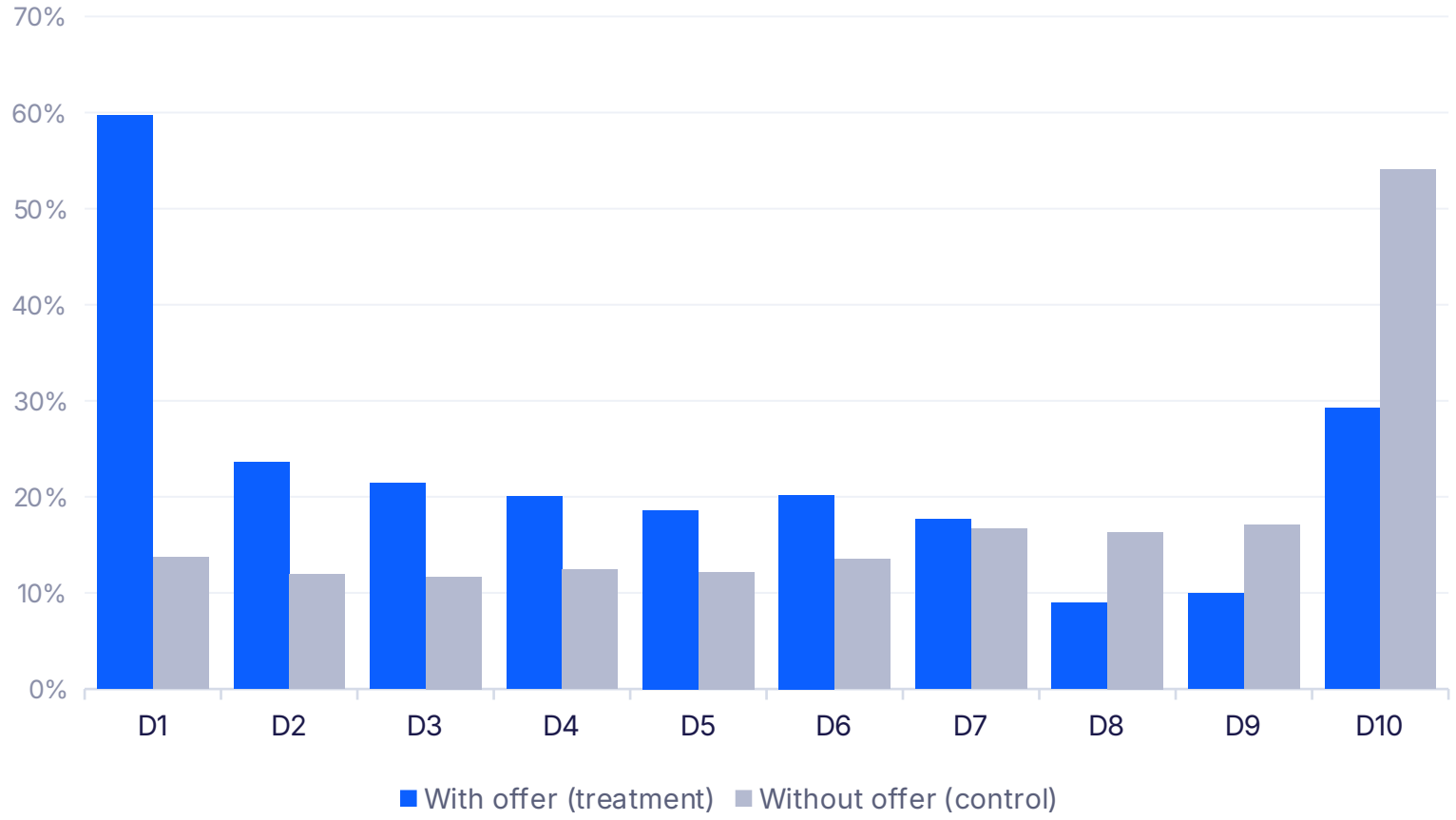
0.0026 SD of its Qini, the smallest of all models

T-learner selected as the final model

Each column is a different random seed for the split; every model re-trained with the tuned settings; SD = standard deviation over the 5 splits

OBSERVED UPLIFT BY DECILE

Test customers sorted by the T-learner's score and cut into 10 groups of 18,000 (D1 = highest)



	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
UPLIFT	+0.46	+0.12	+0.10	+0.08	+0.06	+0.07	+0.01	−0.07	−0.07	−0.25

+46 pp

D1, persuadables

conversion 13.7% without and 59.7% with the offer

−25 pp

D10, sleeping dogs

conversion 54.1% without and only 29.3% with the offer

Observed uplift = treatment conversion rate − control conversion rate inside each decile; held-out test set

CAMPAIGN EFFICIENCY

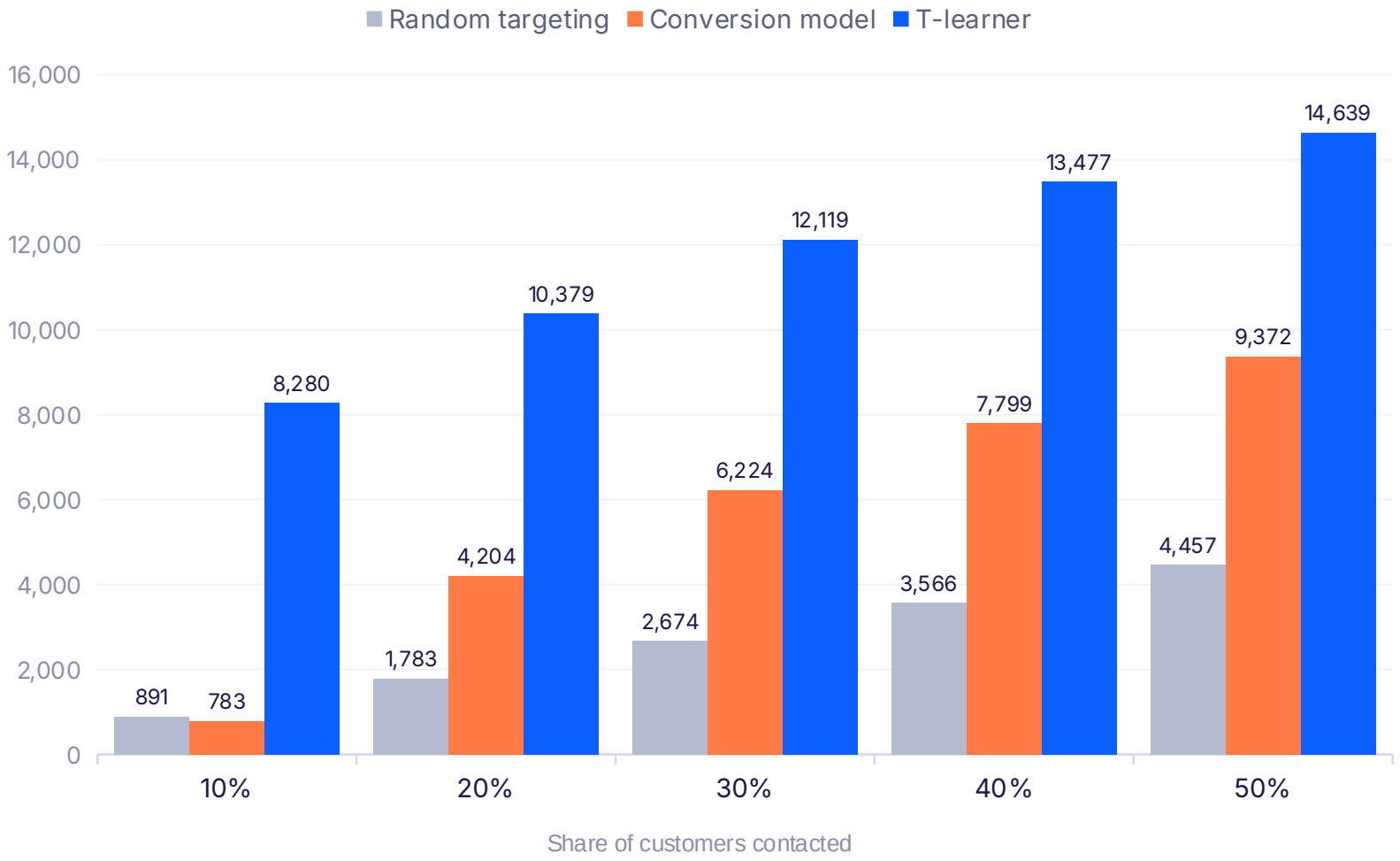
Estimated extra conversions when a share of the 180,000 test customers is contacted

AT A 30% BUDGET

4.5×

more extra conversions than random targeting

TARGETING	EXTRA CONV.
T-learner	**12,119**
Conversion model	6,224
Random	2,674



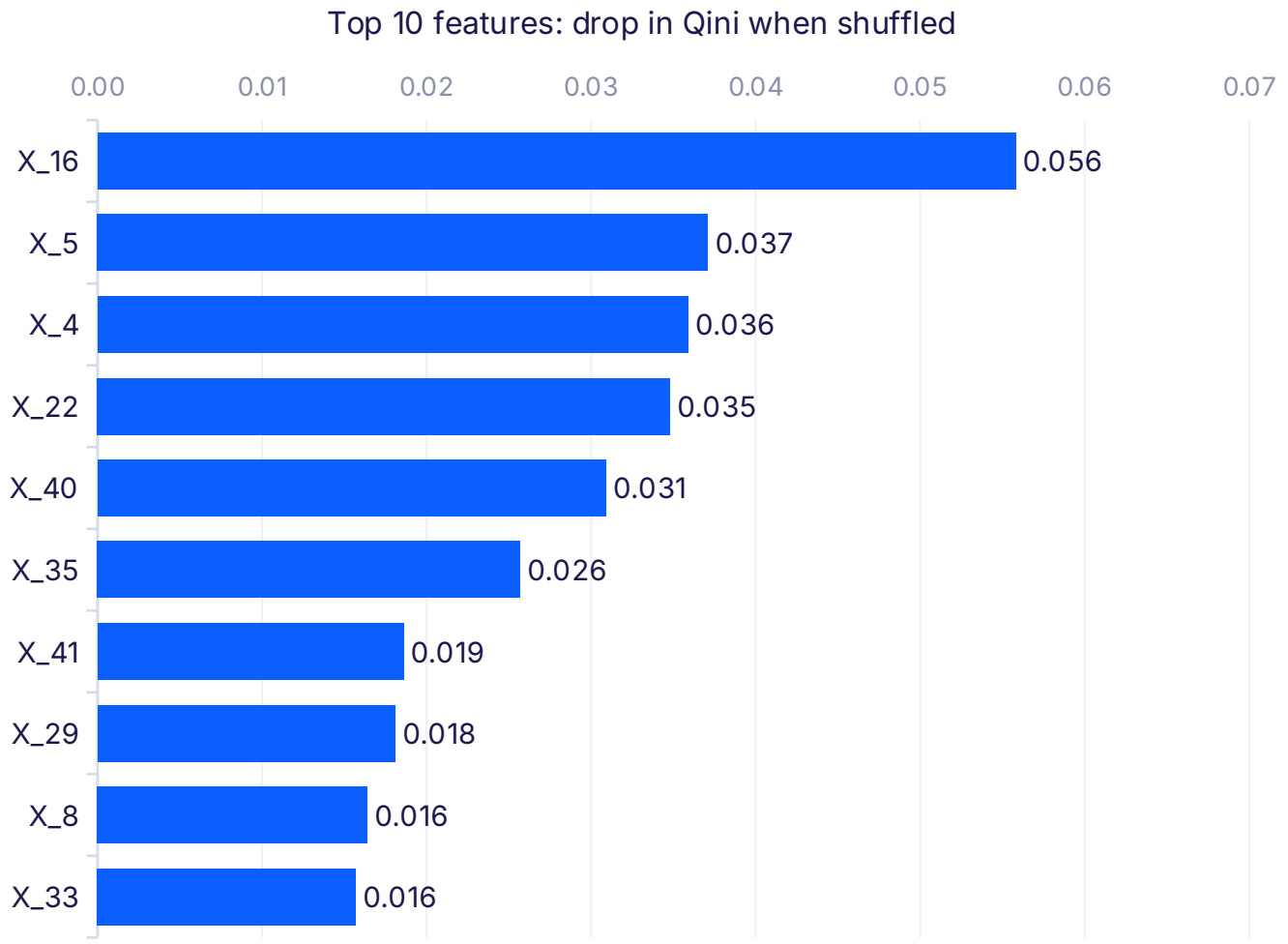
Extra conversions = customers contacted × observed uplift in the selected group; random targeting = customers contacted × 4.95%

NEGATIVE UPLIFT AND FEATURE INFLUENCE

35.2%
of test customers have negative predicted uplift

-11.5 pp
observed effect of the offer in this group

+13.8 pp
observed effect for all other customers



Permutation importance: drop in Qini when one feature is shuffled (60,000 test customers, 2 repeats); features are anonymized, so no business meaning is given

DISCUSSION

Why the T-learner won

- About 210,000 training rows per group is enough for two separate models.
- The S-learner has to learn the effect through one binary feature and can shrink it toward zero.
- The linear T-learner (Qini 0.041) shows the effect is non-linear.

Overfitting and underfitting

- Tuning used only training folds; the test set was untouched.
- min_samples_leaf 100 and L2 1.0 limit tree complexity.
- Qini SD below 0.005 on 5 splits for all boosted models. Only the linear model underfits.

Limitations and improvements

- Synthetic and anonymized data: effects may be stronger than in real campaigns, and features cannot be explained.
- One offer, binary outcome, same cost for every contact.
- Next: X-learner, causal forests, profit-based targeting.

Künzel et al. (2019) on S- and T-learners; results from the notebook

CONCLUSION

1 Target on uplift, not on the chance to buy

T-learner Qini 0.245 against 0.096 for the conversion model, positive on 5 of 5 splits.

2 Same budget, more effect

The top 30% by uplift show 22.4% observed uplift against 5.0% on average, about 4.5 times random targeting.

3 Know whom not to contact

35.2% of customers are expected to buy less if they get the offer.

Practical use

Any operator with a random control group can run this pipeline for upsell, cross-sell or retention offers and send them only to customers with positive uplift.

Future work

- X-learner, R-learner, causal forests
- Include contact cost and revenue
- Several offers at once
- A/B test of the list on a real campaign

All numbers from the held-out test set (180,000 customers)

REFERENCES

Ascarza, E. (2018). Retention futility: Targeting high-risk customers might be ineffective. Journal of Marketing Research, 55(1), 80–98.

Athey, S., & Imbens, G. (2016). Recursive partitioning for heterogeneous causal effects. PNAS, 113(27), 7353–7360.

Devriendt, F., Moldovan, D., & Verbeke, W. (2018). A literature survey and experimental evaluation of the state-of-the-art in uplift modeling. Big Data, 6(1), 13–41.

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. The Annals of Statistics, 29(5), 1189–1232.

Gutierrez, P., & Gérardy, J.-Y. (2017). Causal inference and uplift modelling: A review of the literature. PMLR, 67, 1–13.

Jaskowski, M., & Jaroszewicz, S. (2012). Uplift modeling for clinical trial data. ICML 2012 Workshop on Clinical Data Analysis.

Künzel, S. R., Sekhon, J. S., Bickel, P. J., & Yu, B. (2019). Metalearners for estimating heterogeneous treatment effects using machine learning. PNAS, 116(10), 4156–4165.

MegaFon. (2021). MegaFon Uplift Competition: Data [Data set]. ODS.ai. https://ods.ai/tracks/df21-megafon/competitions/megafon-df21-comp/data

Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

Radcliffe, N. J. (2007). Using control groups to target on predicted lift: Building and assessing uplift models. Direct Marketing Analytics Journal, 14–21.

Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. Journal of Educational Psychology, 66(5), 688–701.

Shevchenko, M. (n.d.). scikit-uplift (Version 0.5.1) [Computer software]. https://www.uplift-modeling.com

APA 7th edition; full list in the research paper